

ANUSHKA DAS

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Portfolio | LinkedIn | GitHub

EDUCATION

Techno Main Salt Lake

Information Technology B.Tech

Kolkata, West Bengal, India

Sept'23 - June'27

Degree in Coursework: Computer Architecture, Operating System, DBMS, Data Structure

EXPERIENCE

Member of Google's Women Techmakers

Remote | Jan 2025 - present

- Engage in a global community supporting women in tech through events, learning resources, and networking.

Zetheta Algorithms Private Limited | Blockchain Risk Developer(DeFi)

Remote | Jan 2026 - 1

month

- Engineered a vulnerability scanner for Solidity contracts to detect critical flaws like Reentrancy and Access Control.
- Developed an algorithmic market-making engine using the Avellaneda-Stoikov model to optimize bid-ask spreads and manage inventory risk.

GSSoc'25 | Open-Source Contributor

Remote | July 2025 - Sept 2025

Contributed to open-source projects by fixing bugs and added new feature modules across 4 open-source repositories

CodSoft | Web Development Intern

Remote | Aug 2025 - 1 month

Built 6+ responsive web pages, improving UI accessibility

SKILLS

Programming Languages: Python, C++, Java, JavaScript (ES6+)

Libraries/Frameworks: OpenAI, Flask, React.js, NumPy, Scikit-learn, XGBoost, Pandas, React.js

Tools / Platforms: AWS, Power BI, HTML, CSS, Docker

Databases: MySQL

PROJECTS / OPEN-SOURCE

Federated Fresh: Hybrid AI Orchestration Terminal | Link

FastAPI, Google Gemini 2.5 Flash,

ChromaDB, Python, DuckDuckGo API, CSS3/JS

- Architected a hybrid AI routing engine that orchestrates between 3 distinct data streamslocal ChromaDB retrieval, live web-search sweeps, and direct LLM inferenceoptimizing response accuracy across diverse query intents.
- Reduced local memory overhead by 75% by offloading high-latency embedding calculations to the cloud (text-embedding-004), enabling stable production deployment on constrained 512MB RAM infrastructure.
- Engineered an asynchronous RAG pipeline using FastAPI BackgroundTasks to achieve zero-lag user interface performance during the simultaneous semantic chunking and vector injection of multi-page documents.

DeepfakeDetector: Frequency-Based AI Inference Engine | Link

React, FastAPI, Python, PyTorch,

OpenCV, Docker, 2D FFT, ResNet-18, Tailwind CSS

- Engineered a frequency-domain AI engine utilizing 2D FFT and ResNet-18 (~11.1M params) to identify synthetic fingerprints invisible to 100% of standard spatial-only CNN architectures.
- Achieved <1.8s end-to-end inference latency for high-resolution media by architecting a distributed React/FastAPI/Docker stack across 2 cloud environments (Vercel & Hugging Face).
- Integrated a 3x3 Gaussian Micro-blur filter to neutralize sensor noise, effectively isolating high-frequency spectral artifacts to enhance detection robustness against modern Diffusion-based generators.

Deweathering Engine | [Link](#)

Python, React, FastAPI, OpenCV

- Engineered a 2-tier full-stack React and FastAPI application using Robust Principal Component Analysis (RPCA) to mathematically extract clean text from degraded documents.
- 95% Automated parameter tuning with a custom Adaptive Regularization heuristic, dynamically scaling the penalty parameter (0.4-0.8) based on matrix noise variance.
- Achieved full document restoration in under 2.5 seconds via optimized OpenCV and NumPy operations, deployed across Render and Vercel cloud infrastructure.

Customer Churn Prediction | [Link](#)

Python, Scikit-learn, XGBoost, Streamlit

- Developed and deployed a real-time churn prediction app using Streamlit, achieving a 67% recall and 0.82 AUC on a dataset of 10,000+ customer records.
- Engineered 15+ custom features, including usage frequency and payment patterns, which improved model precision by 12% compared to baseline iterations.
- Optimized inference logic to deliver predictions in under 200ms, enabling instantaneous risk assessment for retention teams.

CERTIFICATIONS

- Software Engineering Job Simulation - **JPMorgan Chase and Co.**
- Data Analytics Job Simulation - Deloitte Australia.
- Train and manage a machine learning model with Azure Machine Learning - **Microsoft**
- Understand data science for Machine Learning - **Microsoft**
- GenAI Powered Data Analytics Job Simulation - Tata.
- Python for Data Science - **IBM**

HONORS & AWARDS

- Python - @HackerRank Gold Level (5 Star)
- Java - @HackerRank Silver Level (3 Star)
- SIH 2025 (Semi-Finalist) - Top Finalist out of 250+ competing teams in the college internal round for the Smart India Hackathon (SIH) 2025
- Active member of Samaritans social club since first year, contributing to community initiatives and collaborative events and occasionally participate in debate and quiz competitions, showcasing strong communication and critical thinking skills.